

Statistical Inference and Modeling

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3. ~~4.~~ Linear Regression

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Simple linear regression

Components:

- Let x be the independent variables aka predictors aka the input features
- Let $\hat{y}$ be the dependent variables aka responses aka the prediction of the targets.
- Let y be the measurements (reference) *noise*

Suppose that the prediction of target is derived from a linear regression model of x :

$$\hat{y} = \beta_0 + \beta_1 x \rightarrow \text{linear parameter between } x \text{ and } \beta.$$

Annotations: bias (pointing to β_0), weight (pointing to β_1)

Then, we can write the residual error ϵ : the error between the linear model and the measurements, as follows:

$$\epsilon = \hat{y} - y = (\beta_0 + \beta_1 x) - y$$

$$\beta_0 + \beta_1 x + \beta_2 x^2 \checkmark$$

$$\beta_0 + \beta_1 x + \beta_0 \beta_1 x^2$$

non linear,

(1)

where $x, y \in \mathcal{R}$ are the observed features and targets.

To solve for β_0 and β_1 , you need to observe two sets of (x, y) that can form two equations that are linearly independent.

Simple linear regression

However, this may not be the case in practice, as generally we prefer that β_0 and β_1 can fit well with **most of our data samples**.

$$\hat{y}_1 = \beta_0 + \beta_1 x_1^1 \approx y_1$$

$$\hat{y}_2 = \beta_0 + \beta_1 x_1^2 \approx y_2$$

$$\vdots = \vdots \quad \quad \quad \vdots$$

$$\hat{y}_m = \beta_0 + \beta_1 x_1^m \approx y_m$$

Let's say we have m data samples, we solve for β^0 and β^1 that minimizes the following **residual sum of square (RSS)**:

$$\text{RSS} = \sum_{j=1}^m (y_j - (\beta_0 + \beta_1 x_1^j))^2. \quad (2)$$

In other words ...

Estimated

m training samples.

Use train set

$$\hat{\beta}^0, \hat{\beta}^1 = \underset{\beta^0, \beta^1}{\text{argmin}} \sum_{j=1}^m (y_j - (\beta_0 + \beta_1 x_1^j))^2 \quad \text{objective.} \quad (3)$$

Extension to multiple linear regression

- **Multiple linear regression (MLR)** is a statistical technique that uses **several independent variables** to predict the outcome:

$$\begin{aligned}\hat{y} &= \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n \\ &= \beta_0 + \mathbf{x}^T \boldsymbol{\beta}\end{aligned}$$

Handwritten annotations: "bias" above β_0 , "weight." above the $\beta_i x_i$ terms, and blue circles around x_1, x_2, x_n .

The above model can be simplified by assigning zero bias $\beta_0 = 0$. **Several independent**

variables can be employed as n -dimensional input feature $[1]$, i.e., $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathcal{R}^n$ (as

oppose to $x \in \mathcal{R}$)

Matrix format of linear regression

- **Simple linear regression.** Previous linear regression can be expressed as follows (note that we ignore the number of measurements):

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_m \end{bmatrix} = \begin{bmatrix} 1 & x_1^1 \\ 1 & x_1^2 \\ \vdots & \vdots \\ 1 & x_1^m \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \mathbf{X}\boldsymbol{\beta}$$

dummy (pointing to the column of ones)

where $\mathbf{X} \in \mathcal{R}^{m \times 2}$. This simple linear regression uses only **one independent variable** $x \in \mathcal{R}$.

- **Multiple linear regression:** to learn $\{\beta_i\}_{i=1}^n$, we need some m measurements, i.e., $\{(y_j, \mathbf{x}^j)\}_{j=1}^m$ where each input feature $\mathbf{x}^j \in \mathcal{R}^n$. We can form a matrix equation as follows:

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_m \end{bmatrix} = \begin{bmatrix} 1 & x_1^1 & x_2^1 & \dots & x_n^1 \\ 1 & x_1^2 & x_2^2 & \dots & x_n^2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_1^m & x_2^m & \dots & x_n^m \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix} = \mathbf{X}\boldsymbol{\beta}$$

$\chi_i^j \rightarrow$ sample j
 $\chi_i^j \rightarrow$ feature i

$(\chi^1)^T$ (pointing to the first row)

Normally, it is assumed that $\beta_0 = 0$. The regressor matrix reduce to just a set of m input features $\{\mathbf{x}^j\}_{j=1}^m$ and $\mathbf{X} \in \mathcal{R}^{m \times n}$:

$$\mathbf{X} = \begin{bmatrix} x_1^1 & x_2^1 & \dots & x_n^1 \\ \vdots & \vdots & \vdots & \vdots \\ x_1^m & x_2^m & \dots & x_n^m \end{bmatrix} = \begin{bmatrix} (\mathbf{x}^1)^T \\ \vdots \\ (\mathbf{x}^m)^T \end{bmatrix}$$

Problem statement

Settings: we have m measurements, i.e., $\mathbf{y} \in \mathcal{R}^m$ and $\mathbf{X} \in \mathcal{R}^{m \times n}$.

$$\mathbf{y} \approx \hat{\mathbf{y}} = \mathbf{X}\boldsymbol{\beta}$$

Formulation: $\boldsymbol{\beta}$ can be estimated by solving the least-square error minimization:

$$\boldsymbol{\beta}_{ls} = \underset{\boldsymbol{\beta}}{\operatorname{argmax}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 = \sum_j^m (y_j - \sum_i^n x_i^j \beta_i)^2 \quad (4)$$

- $\mathbf{r} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}$ is called *the residual error*.
- $\boldsymbol{\beta}$ that give the smallest $\|\mathbf{r}\|_2$ is called *the least-squares solution*.
- the problem Eq.(4) is equivalent to minimizing $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2$
- $\boldsymbol{\beta}_{ls}$ can be obtained from finding $\boldsymbol{\beta}$ that gives the gradient $\nabla_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 = 0$.

$$\nabla_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}_{ls}\|_2^2 = 2\mathbf{X}^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}_{ls}) = 0$$

$$\boldsymbol{\beta}_{ls} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \dots$$

If $\mathbf{X} \in \mathcal{R}^{m \times n}$ with $m \geq n$ is full rank, then the least-squares solution $\boldsymbol{\beta}_{ls}$ is unique [1].

Pseudo-inverse of $\mathbf{X}$, time complexity $\approx O(n^3)$

Closed-form solution

$$\varepsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$$

$$\hat{\sigma}^2 = \frac{1}{n-1} (y - X\beta_{ls})^T (y - X\beta_{ls})$$

Pseudo Inversion Matrix

$$\beta_{ls} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \dots$$

เป็นการหา $\mathbf{X}^{-1}$ แบบหนึ่ง
ตัวเลขอาจไม่ Exact มากนัก
แต่ช่วยลดปัญหา เมื่อ $\mathbf{X}$ เป็น Singular
หรือคือ $\mathbf{X}_i$ มี dependent กัน.
หรือ $m > n$; full column rank.

$\mathbf{X} \in \mathcal{R}^{m \times n}$ with $m \geq n$ is full rank, which also implies [2]...

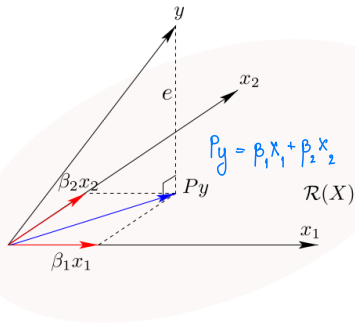
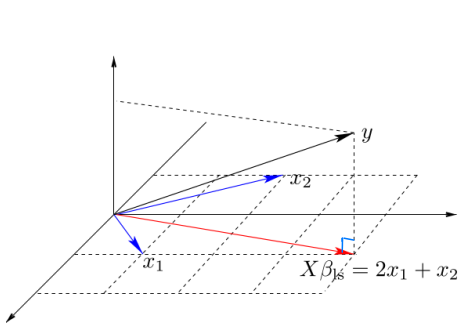
- if $m = n$, then $\mathbf{X}$ is invertible, $(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T = \mathbf{X}^{-1}$. $\rightarrow$ Find the unique solution.
 - Least-squares solution can be found by solving the normal equations...
 - You can perform the matrix inversion using `linalg.inv()`.
- if $m > n$, then $\mathbf{X}$ is non-invertible, but...
 - if $\text{nullity}(\mathbf{X}^T \mathbf{X}) = 0 \Rightarrow \mathbf{X}^T \mathbf{X} \succ 0$. Then, $(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$ can be used as the left inverse of $\mathbf{X}$.
 - If $\mathbf{X}^T \mathbf{X} \succ 0$, you can compute the left inverse of $\mathbf{X}$ with pseudo inversion.
 - Pseudo inversion can be calculated by `linalg.pinv()`

Positive Definite Matrix

- ถ้า $A \succ 0 \rightarrow z^T A z > 0$ เสมอ.
- Convexity.

Pseudo

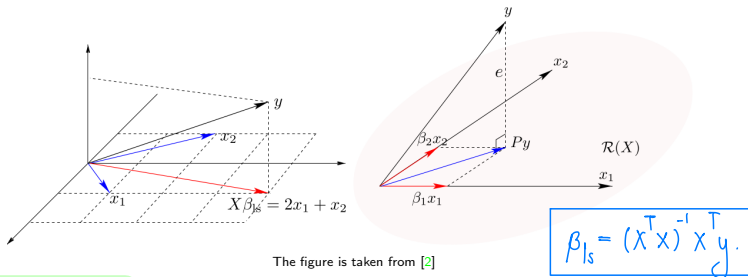
Geometric Interpretation



The figure is taken from [2]

- $\|y - X\beta\|_2$ is the distance from the measurements y to any predictive value $X\beta$
- $\|y - X\beta_{ls}\|_2$ has the shortest distance. (form least square)
- $(y - X\beta_{ls})$ is orthogonal to each column in X ...
... on the other words... $\hat{y} = X\beta_{ls}$ is the orthogonal projection of y to the range of X .

Orthogonal Projection Matrix



- $P = X(X^T X)^{-1} X^T$ is the orthogonal projection matrix (hat matrix).
 - $\hat{y} = X\beta_{ls} = X \cdot (X^T X)^{-1} X^T y = Py$ is the orthogonal projection of y to the range of X .
 - The orthogonal projection matrix has the following properties
 - $P = P^T \rightarrow$ Symmetric.
 - $P^2 = P$
 - $\|P\mathbf{x}\| \leq \|\mathbf{x}\|$ for all $\mathbf{x} \in \mathcal{R}(X)$.
- These properties imply that $I - P \succ 0$
- * need to normalize your input features.

Python: β can be calculated by

- Solvers, e.g., `lstsq`, `LinearRegression`, `sm.OLS`.
- Or you can perform the (pseudo) matrix inversion with `linalg.pinv()`.

Let's try Python coding in [Tutorial 1](#)

7 Performance assessment

Performance on the prediction

Theoretical performamnce — Biasness and MSE

Sampling from distribution — RSS, TSS, R^2

Performance on the modeled coefficients

Theoretical performamnce — Biasness and MSE

Theoretical performamnce — Biasness and MSE

What is the performance of the point estimator $\hat{Y} = f(X) = \mathbf{x}^T \boldsymbol{\beta}_{ls}$?

Note that, we use the measurements y as reference.

We can measure the performance of the point estimator $\hat{Y}$ (or $\hat{Y}_m$) with the followings...

- ① **Bias** $\Rightarrow$ How far is the mean value of Y from the measurements y

$$\text{Bias}_y[\hat{Y}] = E[\hat{Y}] - y = E[\mathbf{x}^T \boldsymbol{\beta}_{ls}] - y$$

- ② **Efficiency** $\Rightarrow$ Suppose that an estimator $\hat{Y}$ is used for estimating y which is a model parameter associated with a probabilistic distribution of a random variable q , or $p(q; y)$. In fact, it is common that the estimator $\hat{Y}$ is also a function of $q_1, q_2, \dots, q_m$ observations that are sampled from $p(q; y)$ and are iid. The efficiency of the estimator $\hat{Y}$ can be defined as:

$$\text{Eff}_y(\Theta) = \frac{LB}{\text{Var}[\Theta]} = \frac{1}{\mathcal{I}_{p(q_1, q_2, \dots, q_m; \theta)}(y) \text{Var}[\Theta]}$$

A good estimator should have the efficiency $\text{Eff}_y(\Theta)$ close to 1 .

- ③ **Mean square error** $\Rightarrow$ It should provide the low theoretical mean square error to the measurements.

$$\text{MSE}(\hat{f}) = E_{X \mathcal{E}}[(f(X) - Y)^2] = \text{Bias}_{y_o}[f(X)]^2 + \text{Var}_X[f(X)] + \sigma^2$$

where the measurements is noisy: $Y = y_o + \mathcal{E}$, and σ^2 is the variance of the noise $\mathcal{E}$.

- ④ **Consistency** $\Rightarrow$ Is the estimator consistent? $\hat{Y}_m$ converges to the measurements y in probability?

$$\lim_{m \rightarrow \infty} P(|\hat{Y}_m - y| \leq 0)$$

where $\hat{Y}_m = \mathbf{x}^T \boldsymbol{\beta}_{ls}^m$ and $\boldsymbol{\beta}_{ls}^m$ is estimated using m training samples.

Sampling from distribution—RSS, TSS, R^2

Let $\mathbf{x}_j \in \mathcal{R}^n$ denotes a feature vector of the j^{th} measurement.

- Residual sum of square (RSS) is

$$RSS = (y_1 - \hat{y}_1)^2 + \dots + (y_m - \hat{y}_m)^2 = \sum_{j=1}^m (y_j - \mathbf{x}_j^T \boldsymbol{\beta}_{ls})^2$$

- Total sum of square (TSS) measuring the total variance is

$$\begin{aligned} TSS &= \sum_{j=1}^m (y_j - \bar{y})^2 \\ &= \sum_{j=1}^m ((y_j - \hat{y}_j) + (\hat{y}_j - \bar{y}))^2 \\ &= \sum_{j=1}^m \text{RSS} + \sum_{j=1}^m (\hat{y}_j - \bar{y})^2 + 2 \sum_{j=1}^m (y_j - \hat{y}_j)(\hat{y}_j - \bar{y}) \end{aligned}$$

Note that $TSS - RSS \leq TSS$

$$TSS - RSS = \left(\sum_{j=1}^m (y_j - \hat{y}_j)^2 + \sum_{j=1}^m (\hat{y}_j - \bar{y})^2 + 2 \sum_{j=1}^m (y_j - \hat{y}_j)(\hat{y}_j - \bar{y}) \right) - \sum_{j=1}^m (y_j - \hat{y}_j)^2$$

- R^2 is defined as

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

Performance on the modeled coefficients

Before we move on to the next step.

Let us assume that

- $\mathbf{X}$ is a deterministic value matrix.
- $\mathbf{y}$ is the measurement vector of the target labels and is corrupted by noise $\mathcal{E} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$.
- Let β denote the linear coefficients that can correctly map $\mathbf{X}$ to the ground truth label $\mathbf{y}_o$, the relationship between these variable can be defined as:

$$\mathbf{y}_o = \mathbf{X}\beta$$

$$\mathbf{y} = \mathbf{y}_o + \mathcal{E}$$

matrix form.



Assumption เป็นจริง

$\mathbf{y}, \mathbf{X} \rightarrow$ Normalized

Standardization $\mathcal{E} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$

Theoretical performnce — Biasness and MSE:

- **Biasness** is defined as $\text{Bias}_\beta[\beta_{ls}] = \mathbb{E}[\beta_{ls}] - \beta$.

$$\begin{aligned}\mathbb{E}[\beta_{ls}] &= \mathbb{E}[(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}] \\ &= \mathbb{E}[(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T (\mathbf{X}\beta + \mathcal{E})] \\ &= \beta + 0\end{aligned}$$

พวกนี้มันค่าคงที่หมด.



Note that $\mathbb{E}[(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathcal{E}] = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbb{E}[\mathcal{E}] = 0$ because the corrupting noise has zero mean.

Therefore, linear regressor is an unbiased estimator.

Theoretical performamnce — Biasness and MSE

Theoretical MSE is defined as

$$\mathbf{MSE} = \text{Bias}_{\beta}[\beta_{ls}]^2 + \text{cov}(\beta_{ls}).$$

- **Covariance** of β_{ls} is defined as

$$\text{cov}(\beta_{ls}) = E[(\beta_{ls} - E[\beta_{ls}])(\beta_{ls} - E[\beta_{ls}])^T]$$

where $E[\beta_{ls}] = \beta$ and that

$$\begin{aligned}\beta_{ls} - E[\beta_{ls}] &= \beta_{ls} - \beta \\ &= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T (\mathbf{X} \beta + \epsilon) - \beta \\ &= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \epsilon\end{aligned}$$

$$\begin{aligned}\text{cov}(\beta_{ls}) &= E[(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \boldsymbol{\epsilon} \cdot ((\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \boldsymbol{\epsilon})^T] \\ &= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T E[\boldsymbol{\epsilon} \boldsymbol{\epsilon}^T] \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \\ &= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \text{cov}(\boldsymbol{\epsilon}) \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \\ &= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T (\sigma^2 \mathbf{I}) \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \quad \text{Close to identity.} \\ &= \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1} \cancel{\mathbf{X}^T \mathbf{X}} (\mathbf{X}^T \mathbf{X})^{-1} = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}\end{aligned}$$

Fisher information that $\mathbf{y}$ carries about β ?

What is $I(\beta)$, the Fisher information that $\mathbf{y}$ carries about β ?

Recall that $\mathbf{y} = \mathbf{X}\beta + \mathcal{E}$ and $\mathcal{E} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$.

Thus, $\mathbf{y} \sim \mathcal{N}(\mathbf{X}\beta, \sigma^2 \mathbf{I})$...

So, we have to find the the Fisher information that the Gaussian variable $\mathbf{y}$ carries about β .

$$\begin{aligned} I(\beta) &= -\mathbb{E}[\nabla_{\beta}^2 \log p(\mathbf{y}|\mathbf{X}\beta, \sigma^2 \mathbf{I})] \\ &= \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1} \end{aligned}$$

This mean that the $\text{cov}(\beta_{ls}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$ is at the lower bound, i.e.,

$$\text{cov}(\beta_{ls}) = I(\beta) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

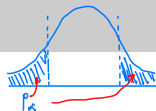
Therefore, the theoretical MSE is at the minimum.

$\beta_{ls} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ is the **Best Linear Unbiased Estimator** for β .

$$\mathbb{E}[\beta_{ls}] = \beta$$

Hypothesis testing of model coefficients

A common question is 'are they likely to be zero?'



↑ Zero model params.
→ Well encode sparse impact features.
(Label input features)
→ Enhancing model complexity (low runtime).

- Let's hypothesize that some estimated coefficients will be zero $\beta_i = 0$, i.e., $H_o = 0$.
- Let's set up $P_\alpha = 0.05$, so $\alpha = -1.65$. (99th Q-function of Normal Distribution)
- Let Z denote a testing statistics. Let's use sample mean in this case:

$$Z = \frac{\sum_{i=1}^n \beta_i}{n}.$$

Suppose that β_i has the variance of s^2 and mean of μ . Suppose that the variance of β_i follows the homoskedastic assumption (i.e., the variance of the residual is constant).

$$\begin{aligned} E[(Z - E[Z])^2] &= E\left[\left(\frac{\sum_{i=1}^n \beta_i}{n} - E\left[\frac{\sum_{i=1}^n \beta_i}{n}\right]\right)^2\right] \\ &= E\left[\left(\frac{\sum_{i=1}^n \beta_i}{n} - \mu\right)^2\right] \\ &= \frac{s^2}{n} \end{aligned}$$

- Then, the normalized version of Z can be calculated from

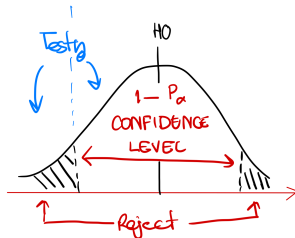
$$W = \frac{Z - H_o}{\sqrt{s^2/n}} = \frac{Z - 0}{\sqrt{s^2/n}}$$

- Now we can check if $W > \alpha$?... If yes, then β_i is likely to be non-zero.

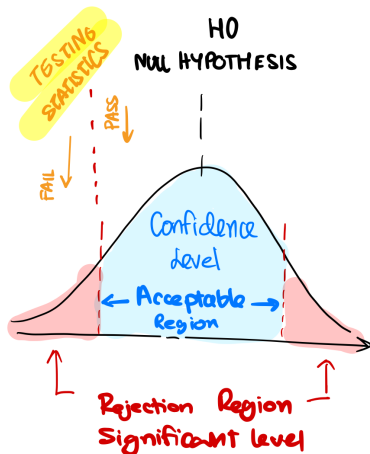
Appendix — Hypothesis testing

General ideas:

- Making an initial assumptions.
 - H_0 : Null hypothesis
 - H_1 : Alternative hypothesis (usually opposite to H_0)
- Collecting evidences (data) to be used in testing the assumptions.
- Based on the testing results, deciding whether or to reject the null hypothesis.



Components

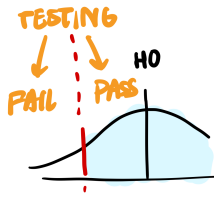


- Null hypothesis
- Acceptance region
- Rejection region
- Critical value
- Testing statistics

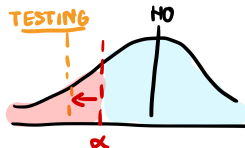
- Acceptance region $\Rightarrow$ Confidence level
- Rejection region $\Rightarrow$ Significant level

Two hypothesis testing approaches

- **Critical value approach.** Finding if the mean of testing samples will cause the rejection of the hypothesis's statistics by comparing it with a **critical value**.

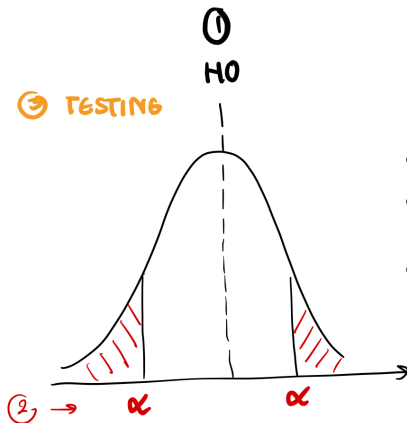


- **P-value approach.** What is the **lowest significant level**/smallest α that will accept the null hypothesis?



Critical value approach — Components

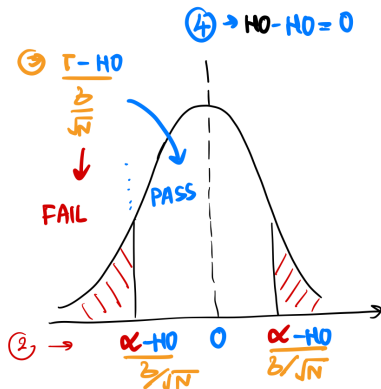
Critical value approach: finding if the mean of testing samples does not violate the hypothesis's acceptable region.



- Set up a null hypothesis.
- Find the critical value. (**significant level** = 1 - confidence level)
- Find the **testing statistics** (mean and variances) of the testing samples.

Critical value approach — Components

Critical value approach: finding if the mean of testing samples does not violate the hypothesis's acceptable region.



- Set up a null hypothesis.
- Define a **significant level** and find the critical value. (**significant level** = 1 - confidence level)
- Find the **testing statistics** (mean and variances) of the testing samples.
- **Normalize the mean with the null hypothesis.**
- Check if it is less than the critical value. If YES, then **Reject**.

Critical value approach — Example I: Student's IQ

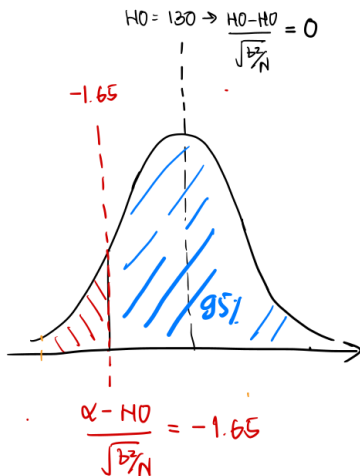
In a class of 20 students, the average IQ is 117, and the std is 10...

Someone told you that Paul has an IQ of 130 or more.

If this statement is given with 95% confidence level, should you reject the above statement?

Note: if this statement is given with 95% confidence level, we can also say that the chance that 'Paul has an IQ of 130' is wrong is about 5%.

Critical value approach — Example I: Student's IQ



- H_0 : Paul has an IQ 130 ($H_0 = 130$)
 - **Let the standard error be 5%.** From Q-table, the area under the normal curve of 5% is at -1.65 . **The critical value = -1.65 .**
 - **The testing statistics.** The average IQ is 117, and std is 10.
 - **Normalize the mean with the null hypothesis.**
- $$W = \frac{\bar{X} - H_0}{\sqrt{\sigma^2/N}} = \frac{117 - 130}{\sqrt{10^2/20}} = -5.81$$
- **Because $-5.81 \leq -1.65 \Rightarrow$ Reject the null hypothesis!**

Example I: Student's IQ — Alternative views

From the standard error $\Rightarrow 5\%$.

The area under the normal curve of 5% is at the value of -1.65 .

We may scale this value up to the testing samples distribution (mean = 130, standard deviation = 10):

$$\begin{aligned}\alpha_{\text{scaled}} &= (-1.65 \times \sqrt{\sigma^2/N}) + H_0 \\ &= -1.65 \times 10/\sqrt{20} + 130 \\ &= 126.31\end{aligned}$$

What does this mean? It means that most of the students are having IQ higher than 126.31!

Note that 126.31 supposes to be the value of an IQ at lowest 5% percentile.

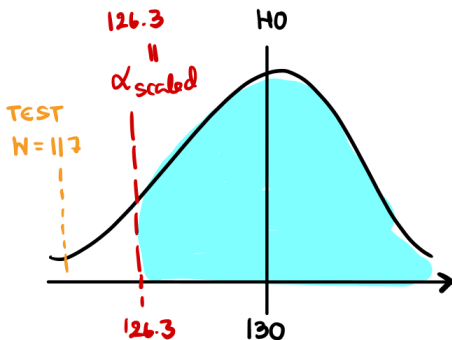
This also means that our hypothesis is suggesting that 95% of students should have the IQ level higher than 126.31.

However, this does not match with the measured value of the average IQ of 117.

Therefore, **we reject the null hypothesis as it is unlikely to happen.**

Example I: Student's IQ — Alternative views

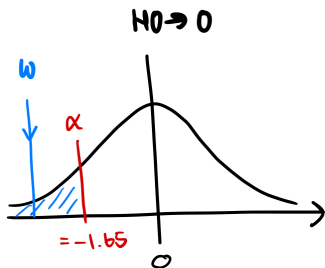
Let's draw the curves....



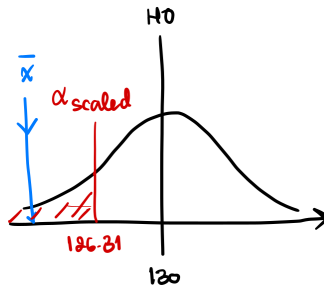
The smaller standard error (or the higher confident level), the less likely that the null hypothesis will be rejected.

Example I: Student's IQ — Comparing two methods

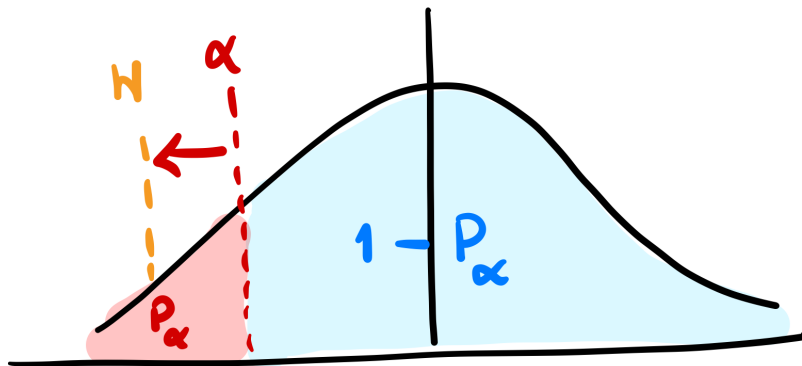
Method 1



Method 2



P-value is the lowest significant level (at the smallest α) that can accept the null hypothesis.



Example — P-Value

What is the **p-value** that can accept Paul's IQ of 126?

P-value is the lowest significant level (at the smallest α):

- $\alpha \leq W = \frac{\bar{X} - H_0}{\sqrt{\sigma^2/N}} \approx -4$
- P value is $Q(-4.0) \approx 0.000031$

Therefore, we reject H_0 for the significant level larger than ≈ 0.000031 . Thus, the P-value is equal to 0.000031

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- [1] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning*. Springer Series in Statistics. New York, NY, USA: Springer New York Inc., 2001.
- [2] Jitkomut Songsiri. “Statistical Learning”. In: in EE575 Random Processes for EE. Chulalongkorn University, 2022. Chap. 2. Linear Regression. URL: <http://jitkomut.eng.chula.ac.th/ee575.html>.